Simple Linear Regression

Importing the libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Importing the dataset

In [2]:
```
dataset = pd.read_csv("01_dataset/Salary_Data.csv")
X = dataset.iloc[:,:-1].values
y = dataset.iloc[:,-1].values
```

Splitting the dataset into the Training set and Test set

In [3]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, r
```

Training the Simple Regression model on the Training set

In [4]:
```
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(X_train, y_train)
```

Out[4]:
```
▾ LinearRegression
LinearRegression()
```

Predicting the Test set results

In [5]:
```
print(X_test)
```

```
[[ 1.5]
 [10.3]
 [ 4.1]
 [ 3.9]
 [ 9.5]
 [ 8.7]]
```

In [7]:
```
print(y_test)
```

```
[ 37731. 122391.  57081.  63218. 116969. 109431.]
```

In [8]:
```
y_pred = regressor.predict(X_test)
```

Visualising the Training set results

In [9]:
```
y_pred
```

Out[9]:
```
array([ 40748.96184072, 122699.62295594,  64961.65717022,  63099.14214487,
       115249.56285456, 107799.50275317])
```

In [10]:
```
plt.scatter(X_train, y_train, color ='red')
plt.plot(X_train, regressor.predict(X_train), color = 'blue')
plt.title('Salary vs Experience (Trainig set)')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.show()
```

Visualising the Test set results

In [11]:
```
plt.scatter(X_test, y_test, color = 'red')
plt.plot(X_train, regressor.predict(X_train), color ='blue')
plt.title('Salary vs Experience (Test set)')
```

```
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.show()
```

this is possible because the dataset had a linear relationship

In [ ]: